Electrical energy

National 5 · Engineering contexts & challenges · Answers

Answers and marking guidance

Working is shown for each answer. Other valid methods should be credited.

1 Calculate the electrical energy used.

- $E_e = 230 \times 10 \times 45 = 103\,500$ J
- = 100 000 J (100 kJ) to two significant figures

2 A 230 V supply charges a battery, drawing 5 A for 2 minutes. Calculate the electrical energy transferred.

- $t = 2 \times 60 = 120$ s
- $E_e = 230 \times 5 \times 120 = 138\,000$ J = 140 kJ

3 A turbine drives a 110 V generator with an output current of 3 A. Calculate the electrical energy produced in 1 hour.

- $t = 60 \times 60 = 3600$ s
- $E_e = 110 \times 3 \times 3600 = 1\,188\,000$ J = 1.2 MJ

4 A 12 V car headlamp draws 4.5 A. Calculate the energy it uses in 20 minutes.

- $t = 20 \times 60 = 1200$ s
- $E_e = 12 \times 4.5 \times 1200 = 64\,800$ J = 65 kJ

5 A 230 V kettle uses 540 kJ of electrical energy in 3 minutes. Calculate the current it draws.

- $t = 180$ s
- $I = E_e \div (Vt) = 540\,000 \div (230 \times 180)$
- $I = 13$ A